

6(a)	$R = \rho L / A$	C1
	$\rho = 0.92 \times 5.3 \times 10^{-7}$ $= 4.9 \times 10^{-7} \Omega \text{ m}$	A1
6(b)(i)	(current =) $6.4 / 1600 = 4.0 \times 10^{-3} \text{ (A)}$	A1
6(b)(ii)	charge = $4.0 \times 10^{-3} \times 3.2 \times 60$ (= 0.768 C)	C1
	number = $0.768 / 1.6 \times 10^{-19}$ $= 4.8 \times 10^{18}$	A1
6(b)(iii)	$6.4 / E = 1600 / (1400 + 1600)$ or $E = 6.4 + (4.0 \times 10^{-3} \times 1400)$ or $E = 4.0 \times 10^{-3} (1600 + 1400)$	C1
	$E = 12 \text{ V}$	A1
6(b)(iv)	$P = I^2 R$ or $P = VI$ or $P = V^2 / R$	C1
	ratio = $(6.4^2 / 1600) / [(4.0 \times 10^{-3})^2 \times 1400]$ $= 1.1$	A1
6(c)(i)	increase	B1
6(c)(ii)	increase	B1
6(c)(iii)	decrease	B1